

Article

Knowledge-Graph-Based Integrated Line Loss Evaluation Management System

Bin Li and Weihuan Wang *

Guangxi Key Laboratory of Power System Optimization and Energy-Saving Technology, School of Electrical Engineering, Guangxi University, Nanning 530004, China; lizhen@gxu.edu.cn

* Correspondence: 18780467963@163.com

Abstract: The loss of power grid efficiency is affected by a number of factors, including the physical characteristics of the equipment used, the operational characteristics of the power grid itself, the characteristics of the electricity being used, and the natural conditions present. The application of a simple line loss rate value is inadequate for the objective evaluation of the implementation of line loss management in different provinces. Accordingly, this paper puts forth an innovative comprehensive evaluation method for municipal power grid line loss management, namely the construction of a comprehensive management evaluation knowledge graph for line loss. This method integrates the dimensionality reduction capability of nuclear principal component analysis with the combined decision support capability of DEMATEL-Entropy Weight-TOPSIS, thereby achieving a comprehensive evaluation of power grid line loss in the whole process and across multiple dimensions. The empirical analysis of cross-sectional data based on line loss management in Guangxi from 2014 to 2016 demonstrates that the line loss evaluation index system proposed in this paper effectively reduces the redundancy of line loss data. To illustrate, the municipal power grid in Guangxi (GL city) has demonstrated a gradual enhancement in the competitiveness of its management and planning loss reduction strategies. However, there has been a notable decline in the competitiveness of its operational and technical loss reduction approaches, with a reduction of approximately 23.7%. Therefore, the overall development of the city's power grid line loss level can be considered stable. This method allows for an objective and reasonable reflection of the actual situation differences in line loss management work, thus providing a basis for future line loss management evaluation.



Citation: Li, B.; Wang, W. Knowledge-Graph-Based Integrated Line Loss Evaluation Management System. *Appl. Sci.* **2024**, *14*, 9462. <https://doi.org/10.3390/app14209462>

Academic Editor: Pierluigi Siano

Received: 4 September 2024

Revised: 29 September 2024

Accepted: 8 October 2024

Published: 16 October 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: municipal grid; knowledge graph; principal component analysis; line loss management; portfolio evaluation

1. Introduction

The decision to achieve a peak in carbon emissions and carbon integration represents a significant strategic choice made by the state with the objective of coordinating the overall situation at home and abroad. Effective line loss management is a crucial aspect of the country's overall development, particularly in relation to the advancement of the power grid. The entire process of electricity flowing from the power plant through transformers, substations, transmission mains and distribution networks, and electrical equipment can be conceptualized as a large-scale AC circuit, comprising a variety of electrical equipment. With regard to the equipment in question, it should be noted that both transmission lines and electrical equipment exhibit a certain degree of resistance. The passage of current through the energy transmission medium gives rise to the generation of heat due to the resistance encountered, thereby resulting in losses. Concurrently, the current in the power transmission network is alternating current, and the iron components in the transformer equipment will be influenced by the alternating electromagnetic field. In the iron core, the alternating electromagnetic field will give rise to eddy currents and hysteresis. It is possible to reduce the energy loss generated by this process by improving the technical standards of the transformer equipment. However, this particular form of loss cannot be avoided.

In terms of technical design, the occurrence of corona phenomena is a common phenomenon near high-voltage transmission lines. This is due to the presence of a number of factors, including the radius of the transmission line cross-section, the density of the air around the transmission line, the presence of aerosols around the transmission line, and the physical properties of the medium around the transmission line. Corona losses have a considerable impact on the normal functioning of the power grid and the economics of grid operation. In the process of selecting equipment, it is possible for capacitive currents to occur between conductors, particularly in the case of high-voltage transmission lines. Concurrently, insulators are subjected to the effects of the electric field, temperature, and environmental variables over extended periods of operation, leading to a decline in insulation efficacy and an augmentation in power loss.

In the context of management planning, power losses resulting from inadequate power dispatching, delayed data collection on power consumption at the border, inaccuracies in power meters, and power theft by users are also classified as line losses. The term “line losses” encompasses both the power loss caused by the equipment itself and the technical management thereof. In accordance with the stipulations set forth by the power grid company, the discrepancy between the settlement value of the on-grid power transmitted from the power plant to the power grid and the settlement value of the user’s power consumption is defined as the power grid line loss. The line loss rate can be defined in two ways: as the ratio of the active power loss of the power grid to the total power transmitted by the line, or as the ratio of the difference between the supply and sales of electricity to the total transmission. To a certain extent, the level of line loss can serve as an indicator of the technical and managerial proficiency of the power company responsible for the operation of the power grid [1]. It can be reasonably deduced that the implementation of grid management has the potential to facilitate technological advancement and innovation within the power industry. Furthermore, it can be expected that this will result in the industry becoming increasingly intelligent, digital, efficient, and environmentally friendly, while simultaneously enhancing its overall competitiveness and core competitiveness [2,3]. At present, the evaluation methods and indicators employed by power grid companies for the management of line loss have disparate foci. In a notable contribution to the field, literature [4] has developed a comprehensive knowledge graph that integrates a vast array of data, topological structures, components, and customer metadata related to the power grid system. The following examples demonstrate the utilization of GridKG for the analysis of the power grid, including the identification of electrical paths and the localization of switches and protective equipment sequences under specific conditions. However, the limitation is that data collection for distribution networks are often singular and subjective, that relevant data are insufficiently interpreted, and that new data are not adequately characterized. Literature [5] proposes a comprehensive framework for quantifying and modeling the resilience of power systems, with a particular focus on the impact of natural disaster events on the system. The authors put forth a novel systematic resilience matrix method based on the axiomatic design concept for the numerical representation of large flexible engineering systems. Nevertheless, the algorithm proposed in the paper may be computationally expensive when applied to large-scale power systems, particularly when performing Monte Carlo simulations and optimal power flow calculations. In order to render the model feasible, it may be necessary to simplify the complexities of the real world. One such simplification would be to assume that all generators remain resilient during wind events, which may not fully reflect the actual situation. The paper [6] puts forth a comprehensive methodology for quantifying the impact of diverse automated fault management schemes on the reliability of distribution systems. The paper puts forth a modular approach to evaluating the impact of automation schemes on system reliability, comprising the breakdown of automation schemes into independently analyzable modules. However, the paper does not present a detailed argument for each refined indicator. For instance, while the paper references the concepts of reliability cost and value, it does not fully explore the long-term economic implications of implementing automation schemes,

including the initial investment, operational costs, and potential savings. Concurrently, distribution systems in disparate regions may exhibit disparate design standards, operating environments, and fault patterns, necessitating adjustments to the methods delineated in the paper to align with the particular circumstances of each region.

With regard to the aforementioned comprehensive evaluation methods, it is evident that the application of automated fault management in power grids or the introduction of a comprehensive framework, comprising four steps (data requirements, disaster modelling, line status and system topology determination, and resilience index evaluation), to quantify and model the resilience of power systems, the selection of indicators is either one-sided, without exploring the characteristics of line loss, focuses on the analysis of distribution network line loss faults, or focuses on data analysis on the transformer side and the user side, lacking consideration of the dimensions of line loss indicators and the extraction of knowledge from different structured data. At present, there is a paucity of comprehensive evaluations of the entirety of the line loss process within the field of power grids. In these papers [7–9], the development of smart grid technology necessitates a combination of computer science, artificial intelligence, and knowledge of specific application fields. The processing of data through neural networks or enhanced recommendation systems is contingent upon the analysis and processing of a substantial quantity of data. As a consequence of the process of urbanization and the aging of power grid facilities, coupled with the irrationality of planning, there is an urgent need to carry out a comprehensive and scientific evaluation of the power grid. The objective of such evaluations is to identify the weak links in the power grid system, thereby providing precise targets and direction guidance for the upgrade, transformation, and planning of the power grid. In light of the accelerated advancement of artificial intelligence technology, the knowledge graph (Knowledge Graph, KG), a nascent cognitive computing methodology, has commenced its deployment in the domain of electric power. This method effectively addresses the complex knowledge structure and large-scale, multi-source, heterogeneous data involved in the power grid evaluation process, while integrating the professional knowledge and experience of domain experts and technical personnel. Accordingly, this paper proposes that the lean management of line loss in electric power enterprises be taken as the point of departure, with the objective of integrating data-driven and artificial intelligence technology to construct a line loss management knowledge graph and integrating multi-source data through the knowledge graph with a view to reducing data redundancy.

The dataset pertaining to the power grid is both extensive and intricate, rendering a straightforward comparison of data insufficient for demonstrating the rationality of the indicators. Consequently, it is imperative to employ mathematical and statistical techniques for the analysis of data, thereby reinforcing the conclusions presented in this paper. The three papers in the literature [10–12] provide further insight through NVW random comparison combination, combined evaluation, or model combination, thereby assisting decision-makers in making more informed decisions. Furthermore, by integrating multiple evaluation indicators and methods, they enhance the model's resilience under diverse conditions and uncertainties. This study introduces a line loss feature indicator screening method based on kernel principal component analysis (KPCA) to address the inherent uncertainty and diversity in power grid indicators. This method is designed to address the inherent complexity and nonlinear relationships present in the data. By mapping the line loss features from the original low-dimensional state space to a higher-dimensional space, it is possible to more effectively reveal the intrinsic connections between feature quantities.

Moreover, this study puts forth a composite evaluation theory that is well-suited to line loss analysis. This paper presents a set of power grid evaluation indicators, constructed based on the DEMATEL-entropy weight-TOPSIS composite evaluation method. This method employs a comprehensive assignment approach, integrating subjective and objective factors. The system is designed to address the current challenges in power grid evaluation, where the available evaluation knowledge is rapidly expanding but remains

heavily reliant on traditional storage formats, such as text. Additionally, the standardization, publicization, and intelligence levels of comprehensive evaluation work are inadequate, which constrains the ability of workers to swiftly master and apply evaluation results.

In conclusion, this study presents a comprehensive evaluation system for line loss based on a knowledge graph. This system not only improves the scientific and accurate evaluation of the power grid but also enhances the efficiency of evaluation work through intelligent methods, thereby providing robust decision support for the sustainable development of the power grid.

This study makes the following contributions to the field:

- A. Construct a knowledge graph for the comprehensive evaluation and management of line loss. In light of the current disarray surrounding line loss indicators and the dearth of multi-dimensional indicators, a heterogeneous knowledge map pertaining to line loss has been constructed. By means of knowledge mining in the knowledge graph, the interconnections between disparate bodies of knowledge are identified, thus facilitating a comprehensive evaluation and management of line loss. By mining knowledge within the comprehensive evaluation and management knowledge graph for line loss, the heterogeneous knowledge pertaining to line loss is structured and the associations between knowledge are established. The graph describes and classifies line loss characteristics through the use of special attributes, thereby addressing the issue of the chaotic screening of line loss characteristics and facilitating the selection and business application of line loss characteristics.
- B. Index screening based on principal component analysis. In the process of index extraction, this paper employs the method of principal component screening, which reduces the correlation between indicators through dimensionality reduction, reduces the number of indicators to be analyzed, while ensuring that as much as possible of the original indicator information is retained, thus enabling a comprehensive analysis of the screened indicator data.
- C. A synthesis of multiple evaluation methods. In the evaluation, a combination evaluation theory is proposed that is suitable for the characteristics of line loss analysis. The index evaluation system is constructed on the basis of the DEMATEL-entropy weight-TOPSIS combination evaluation method, with a focus on the DEMATEL subjective valuation method and the entropy weight objective valuation method. The discrepancies in the project ranking, as determined by the application of disparate methodologies, in addition to the associated ranking scores, are presented.

2. Constructing a Knowledge Graph for Comprehensive Line Loss Evaluation Management

In order to address the issues associated with line loss indicators, it is essential to undertake two key steps: knowledge acquisition and knowledge extraction within the knowledge graph. In the field of power systems, which are characterized by complex knowledge systems, researchers have recently begun to explore the potential of knowledge graphs as a means of visualizing knowledge and enhancing the efficiency with which knowledge is utilized. In their paper, Rui Liu [13] and others proposed multiple data processing and demonstrated the knowledge learning and mapping construction process. However, they did not indicate the role of the knowledge graph in the reference scheme or its use as a decision-making tool to improve the operational efficiency of the power system. Similarly, Chen Qian et al. [14] proposed the role of KG in the fault diagnosis of the power system with the aim of improving the operational efficiency of the power system. Chen Qian et al. [15] conducted a systematic review and summary of the application of KG in power system fault diagnosis and processing, with the aim of providing sufficient and comprehensive guidance for further research in this field. Chen Junbin [16] reviewed the knowledge graph technology from the perspectives of knowledge extraction, knowledge representation, knowledge mining, knowledge reasoning, knowledge fusion, and the application of knowledge graphs. The review included examples of vertical risk control,

operation mode analysis, optimization models, and overshooting parameters. However, the study lacked a correlation test of the data.

A review of existing research findings reveals that, although the technical architectures of domain knowledge mapping construction employed by different research teams exhibit slight differences, the core modules encompass knowledge extraction, knowledge fusion, knowledge processing, and other components [17]. Accordingly, this paper employs the entity information extraction method of principal component analysis to perform data dimensionality reduction, feature extraction, data compression, and denoising on a substantial number of data indicators. A comprehensive evaluation of the entity mapping is conducted to characterize the actual grid network topology planning line loss, as well as the management, operation, technology, and other line losses. This is employed to facilitate entity query and matching services for the evaluation logic mapping.

The index elements identified through principal component analysis inform the generation of evaluation logic maps. The evaluation logic map, as defined in this paper, represents a networked and structured expression of the specific evaluation logic within the traditional evaluation system. It places emphasis on the evaluation indexes, index weights, comprehensive evaluation criteria, and other relevant elements. The generation of the evaluation logic map is guided by the evaluation elements defined in the data model. This allows the application of natural language processing, knowledge extraction, information structuring and other technological means, such as the definition of the evaluation indexes and the method of calculating the indexes. Furthermore, the application of natural language processing, knowledge extraction, information structuring, and other technical means allows the specific contents, such as the definition of evaluation indexes, the calculation method of indexes, the design process of index weights, and the evaluation criteria model, to be obtained.

Figure 1 depicts the methodology employed in the construction of the comprehensive evaluation management knowledge graph for line loss.

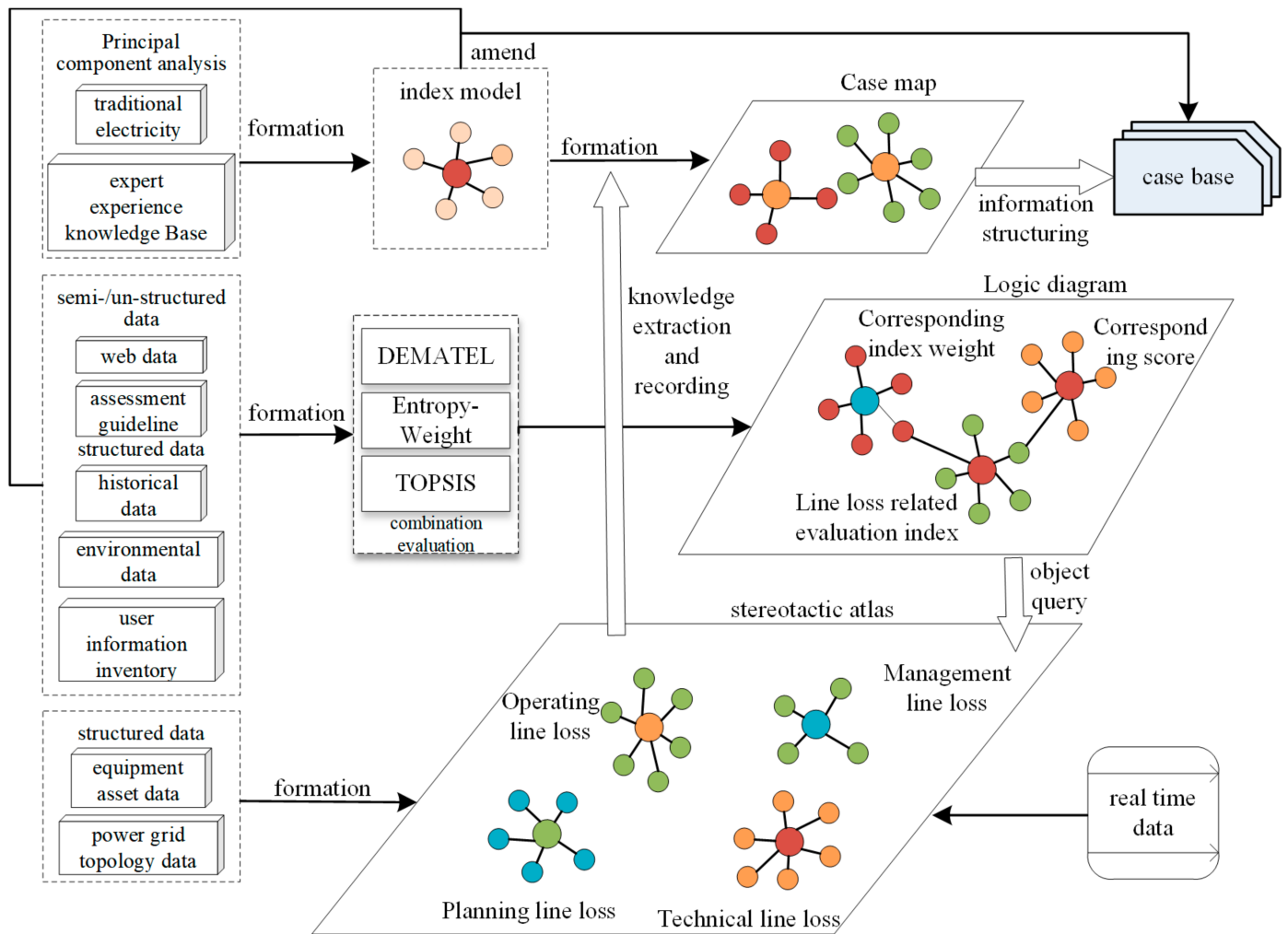


Figure 1. Knowledge graph of line loss comprehensive evaluation management.

3. Calculation and Analysis of Data in the Comprehensive Evaluation of Line Loss Management Entity Mapping

3.1. Configuring Line Loss Management Evaluation Indicator Data Sources

During the process of electricity transmission, electrical energy loss is caused by a number of factors, including resistance and inductance. This type of loss is an unavoidable consequence of the operation of the power system. Nevertheless, through a comprehensive understanding of the underlying causes and evolution of this phenomenon, it is possible to implement effective strategies to mitigate losses and enhance energy efficiency. Line loss is a pervasive phenomenon that affects the entire value chain of the power system. Reducing the line loss rate is a fundamental objective in the pursuit of comprehensive cost reduction and efficiency enhancement for power grid enterprises. It is also an indispensable prerequisite for attaining the “dual carbon” objectives [18]. In accordance with the stipulations set forth in the “Line Loss Management Measures” of the China Southern Power Grid, this paper seeks to collaborate with power grid enterprises in leveraging efficacious information technology tools and computational methodologies to procure, process, and analyze substantial data pertaining to the characteristics and behaviors of power grids. In light of the findings from the literature review on various indicator systems, a knowledge graph is employed to extract and construct an indicator dataset for the line loss management indicator system. In order to obtain multi-time series big data related to line loss in electricity production, this project employs distributed computing technology to collect and analyze data from three principal platforms: the power grid operation and equipment monitoring platform, the power enterprise marketing platform, and the power

enterprise management platform [19]. The data types can be broadly classified into four categories, as illustrated below:

The data set comprises information pertaining to the equipment in question. The model and rated voltage of lines, transformers, other AC/DC components, and other objects under normal operating conditions, as well as other equipment parameter information, are to be included in the model.

Operational data comprises the following: information pertaining to operational parameters, including the voltage, current, and power, is also collected for various electrical components and objects under normal operating conditions.

Capacity data comprise the following: the capacity data of lines, transformers, other AC/DC components, and other objects under normal operating conditions is to be provided on an hourly, daily, or monthly basis. Additionally, information on the quantity and types of electricity meters is to be included.

Topological data may be defined as the information pertaining to the configuration and connectivity of a given system. Information pertaining to the topological connections of lines, transformers, other AC/DC components, and other objects under normal operating conditions.

3.2. Extraction of Relevant Data Indicators

A comprehensive evaluation is a method of assessment that is widely employed in academic contexts. It is an objective, impartial, and logical overall evaluation of the subject under examination, with the aim of accurately positioning the subject. Holistic evaluation is comprised of five elements: the subject of evaluation, evaluation indicators, weight factors, composition models, and evaluators. In order to address the differences in data structures, this paper is based on a standardized processing procedure and method for multi-source heterogeneous data. The indicators are extracted in accordance with five principles.

Systematicity is defined as follows: it is essential to give full consideration to the overarching strategic objectives of the network company, to ensure a clear logical correlation between the indicators, and to facilitate the effective implementation of strategies.

The evaluation must be comprehensive. It is essential to integrate the pertinent elements of line loss management in order to facilitate comprehensive optimization.

Structurality: in accordance with the data, the indicators should be decomposed into structures at the strategic and management levels, with each level providing support to the other.

The objective of differentiation is as follows: in conjunction with the actual characteristics of the network company, a purposeful indicator system should be established.

Correlation: it is recommended that key indicators be selected, related indicators be determined, and collaborative management among different network departments be promoted.

In the data samples obtained, some data are of an unreasonable format and quality due to the presence of errors resulting from manual meter reading and the use of abnormal collection equipment. Consequently, prior to the processing of the data, certain data points are excluded on the basis of their attributes and the line loss characteristics that they reflect.

3.3. Data Processing and Planning Unification

This type of scoring principle is applicable to indicators for which a basis and reference standard for scoring have been established. The evaluation is based on the relevant design guidelines and regulations of the municipal network enterprise, etc. Different scores are set for different indicator data intervals. In the case of indicators that are either very large or very small, the value of each indicator is determined according to the scoring basis or reference standard, with the largest or smallest value being given priority.

For very large indicators:

$$x'_{ij} = \frac{x_{ij} - \min(x_{1j}, x_{2j}, \dots, x_{nj})}{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - \min(x_{1j}, x_{2j}, \dots, x_{nj})} \quad (1)$$

where x'_{ij} denotes the normalized value of the j th positive indicator of the i th evaluation object, x_{ij} is the original data, $\max(x_{1j}, x_{2j}, \dots, x_{nj})$ is the maximum value of the j th indicator among all evaluation objects, and $\min(x_{1j}, x_{2j}, \dots, x_{nj})$ is the minimum value of the j th indicator among all evaluation objects.

For very small indicators:

$$x'_{ij} = \frac{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - x_{ij}}{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - \min(x_{1j}, x_{2j}, \dots, x_{nj})} \quad (2)$$

where x'_{ij} denotes the normalized value of the j th positive indicator of the i th evaluation object, x_{ij} is the original data, $\max(x_{1j}, x_{2j}, \dots, x_{nj})$ is the maximum value of the j th indicator among all evaluation objects, and $\min(x_{1j}, x_{2j}, \dots, x_{nj})$ is the minimum value of the j th indicator among all evaluation objects.

3.4. Expert Integer Scores for the DEMATEL Method

This study incorporates the perspectives of numerous scholars, utilizing the line loss management index system developed in pertinent literature to guarantee the comprehensiveness and integrity of the preliminary set. In order to construct the risk assessment influencing factors survey for line loss management in this study, a Delphi method was employed, based on the evaluation dimensions and index selection principles. The construction of the index is based on the following steps:

- A. Experts with knowledge and experience in related fields were selected, primarily comprising power grid engineers, operation specialists from municipal network companies, and university teachers.
- B. Delphi Survey: The experts were provided with background materials on the indicators, which included a list of potential evaluation indicators, such as equipment performance, operational efficiency, planning strategies, and so forth. They were then asked to score the importance of each indicator. Additionally, experts were permitted to propose indicators they deemed important but not listed. A survey form was distributed to solicit expert opinions anonymously.
- C. Expert opinions were analyzed and summarized to obtain the results of the first round of indicator surveys.
- D. Following a series of anonymous consultations and feedback sessions, the final analysis was presented.

The number of dimension indicators is set to be approximately equivalent, which effectively reduces differences and allows for a comparison between municipal network companies. The correlation between indicator factors is quantified using a 0–4 scale method. The evaluation scale is set as follows: a value of 0 indicates that there is no discernible impact between the two indicators. A value of 1 implies a minor influence between the two indicators. A value of 2 signifies an average influence between the two indicators. A value of 3 denotes a notable influence between the two indicators, while a value of 4 represents a pronounced influence between the two indicators. Based on the results of the experts'

assignments, an initial direct impact matrix M was established, as follows: x_{ij} indicates the degree of impact of indicator F_i on indicator F_j .

$$M = \begin{bmatrix} 0 & x_{12} & \cdots & x_{1n} \\ x_{21} & 0 & \cdots & x_{2n} \\ \vdots & \vdots & 0 & \vdots \\ x_{n1} & x_{n2} & \cdots & 0 \end{bmatrix} \quad (3)$$

3.5. Building a Line Loss Indicator Model Based on Kernel Principal Component Analysis

The loss of electric energy can be attributed to a number of factors, including transmission, transformation, distribution, and electricity consumption. However, the complex topology and diverse forms of anomalous line losses give rise to ambiguity regarding line loss indicators. Consequently, it is of paramount importance to minimize the burden of data collection, enhance the independence of the selected indicators, and augment the data information embedded in the individual indicators.

In light of the aforementioned requirements, this paper proposes the selection of line loss feature indicators based on kernel principal component analysis (KPCA). This approach maps the line loss feature quantities from a low-dimensional state space to a high-dimensional space, thereby facilitating a more comprehensive exploration of the relationships that exist between the feature quantities. In the process of knowledge extraction and knowledge fusion within the knowledge graph, an analysis of the factors affecting the line loss rate for each indicator is conducted on the basis of the aforementioned analysis in the text. The kernel principal component analysis method is employed for the reduction of the dimensionality of the indicators [20]. Subsequently, a composite evaluation method is adopted for the construction of the evaluation process, which subsequently calculates the classification evaluation prediction accuracy of the evaluation indicator system in the evaluation. This enables the influence coefficients of each indicator under this method to be obtained.

Principal component analysis (PCA) is a significant statistical method that can be employed to transform a multi-indicator problem into a smaller number of composite indicators [21]. For a sample space of order $N \times P$ containing N samples, each containing P -dimensional indicators, the matrix is represented as

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{Np} \end{pmatrix}$$

The cumulative variance contribution of the first n principal components is

$$\rho = \sum_{i=1}^n \lambda_i / \sum_{j=1}^P \lambda_j \quad (4)$$

When $\rho \geq 0.70$, it can be assumed that the first n principal components cover most of the information in the original data, and the contribution rate of each principal component is used as the weight, which constitutes the comprehensive evaluation index as

$$f = \alpha_1 y_1 + \alpha_2 y_2 + \cdots + \alpha_n y_n \quad (5)$$

where n is the number of adopted principal components; α_j is the contribution rate of the j th principal component. The higher the comprehensive line loss index, the higher the base line loss rate in the urban area due to various objective factors, such as the grid structure, equipment condition, load level, etc., which should be taken into account when evaluating the level of line loss management.

Figure 2 illustrates the construction of the line loss indicator system based on principal component analysis. This method transforms multiple indicators into multiple multivariate statistical composite indicators, while ensuring that the line loss information is screened less. Once screening has been completed, the comprehensive indicators are referred to as principal components. Each principal component is a linear combination of the original variables, and each principal component is uncorrelated with the others. This makes the principal components perform better than the original variables.

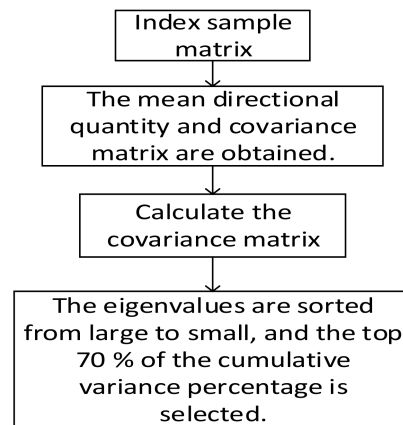


Figure 2. Construction of line loss index system based on principal component analysis.

In addressing problems that are not linear in nature, the use of PCA, which employs linear transformations, often results in suboptimal outcomes. Consequently, scholars have proposed a nonlinear dimensionality reduction method that involves projecting data into a higher-dimensional space before performing PCA operations, which is known as Kernel Principal Component Analysis (KPCA). Kernel principal component analysis (KPCA) maps the original data into a higher-dimensional space using a kernel function and then performs PCA dimensionality reduction within this space, proving an effective approach for handling nonlinear issues. In this paper, the kernel matrix K represented by $N \times N$ is defined as follows:

$$K_{ij} = \langle \varphi(x_i), \varphi(x_j) \rangle \quad (6)$$

By incorporating Equation (6), the inner product can be obtained. Given the high degree of uncertainty and nonlinearity in the management of power grid line losses, this paper selects the Gaussian kernel function as the kernel function for the KPCA algorithm through experimental determination, with the expression given by

$$K(x, x_i) = \exp\left(-\frac{\|x - x_i\|^2}{2\sigma^2}\right) \quad (7)$$

where x and x_i are data points in the original space, and σ is the bandwidth parameter of the Gaussian kernel. For a new set of observational data, it is projected into the feature space, and the feature vectors $v_j (j = 1, \dots, p)$ are extracted (where p represents the number of principal components to be extracted), resulting in the principal components:

$$t_j = \langle v_j, \varphi(x) \rangle = \sum_{i=1}^N \alpha_i^j \langle \varphi(x), \varphi(x_i) \rangle \quad (8)$$

Kernel principal component analysis (KPCA) is an algorithm that performs dimensionality reduction by solving an eigenvalue problem for a kernel matrix, without the need to explicitly consider the nonlinear mapping relationship φ . To validate the superiority of the dimensionality reduction method presented in this paper, traditional principal component analysis (PCA) and kernel principal component analysis (KPCA) were applied to the same

set of line loss indicator original sequences for dimensionality reduction. The effects of high-dimensional data are shown in Figures 3 and 4.

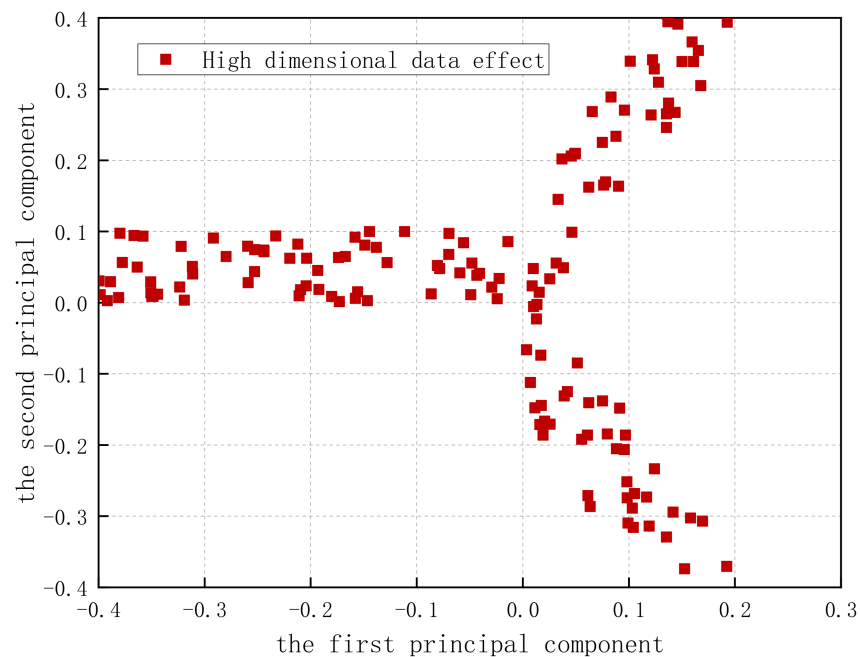


Figure 3. Visualization of dimensionality reduction with kernel PCA (KPCA).

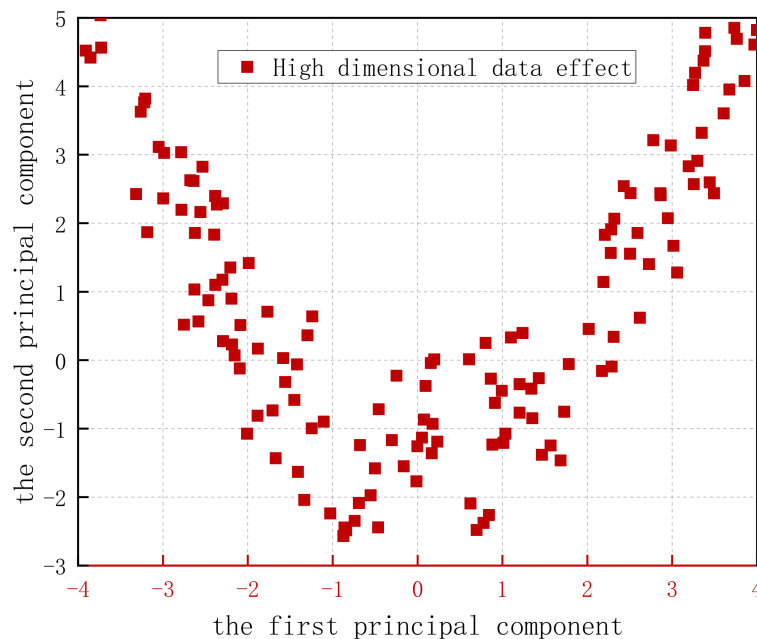


Figure 4. Visualization of dimensionality reduction with principal component analysis (PCA).

4. Construction of Line Loss Comprehensive Evaluation Management Logic Diagram

The kernel principal component analysis has the effect of screening out the line loss indicator system, with the result that the indicators have their own focus. However, there is a possibility that the indicators may be unable to cope with the actual municipal power grid line loss management system. In such cases, the practical application of the indicators often needs to be assigned in order to reflect the objectivity of the indicator evaluation results. The evaluation of indicators employs a variety of methodologies, with some leaning towards expert subjectivity and others tending towards objectivity. Given the multifaceted

nature of existing line loss management, this paper proposes a hybrid evaluation approach, drawing upon the knowledge map of line loss indicators [22,23].

In light of the aforementioned challenges, this paper proposes an integrated approach that combines the single evaluation method of line loss indicators with the evaluation ideas between methods, thereby achieving an “organic combination”. The integration of this method is evident in various aspects of the evaluation method for local conditions, which may be described as a process of “re-engineering”. The study of line loss evaluation object indicators can be broadly divided into four dimensions: planning, management, operation, and technical. It is possible that the evaluation results of different dimensions may appear as extreme values and outliers, which could result in a certain degree of deviation in the overall evaluation results of the indicators. Accordingly, the portfolio evaluation presented in this paper is based on the DEMATEL subjective assignment method and the Entropy-Weight and TOPSIS objective assignment method, which collectively serve to construct the basic framework for portfolio evaluation. Figure 5 is the technical roadmap diagram, which illustrates the design of the technical aspects in the actual application of this project.

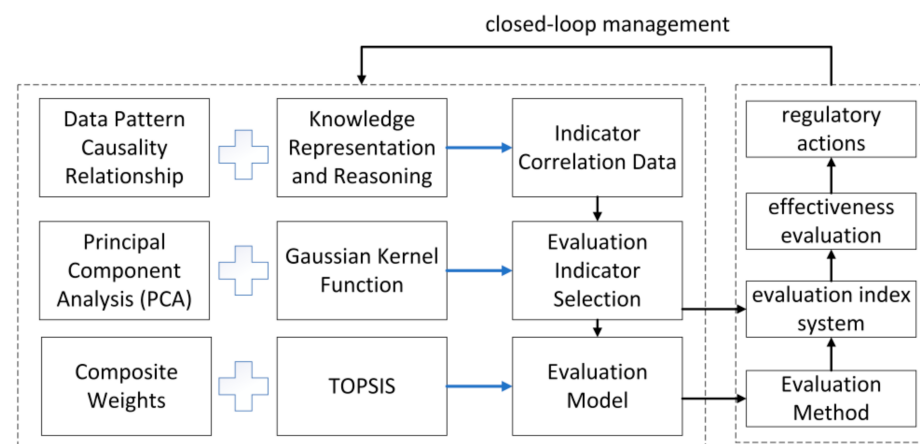


Figure 5. Technical roadmap diagram.

4.1. Fuzzy DEMATEL Method

The DEMATEL (Decision-Making Trial and Evaluation Laboratory) method, also known as the Decision-Making Trial and Evaluation Laboratory method, is a system analysis technique that employs graph theory and matrices. It was first proposed by Gabus and Fontela in the United States in 2010. The method was first proposed by Gabus and Fontela in 2010. The triangular fuzzy numbers introduced in fuzzy set theory serve to enhance the DEMATEL method, circumventing the potential subjectivity of expert input, quantifying the correlation between factors, addressing the ambiguity inherent in the evaluation process, and ensuring the accuracy and reliability of the resulting weightings [24]. By analyzing the logical relationship between each element in the system and the direct influence relationship between the phases, the degree of influence of each factor on other factors and the degree of being influenced is calculated. This allows the cause degree and center degree of the factors to be calculated [25].

Literature [26] calculated the influence degree D_i , the influenced degree C_i , and the centrality degree M_i and the cause degree R_i for each element from the comprehensive influence matrix Y . The centrality degree M_i represents the importance of the indicator in the system. The centrality M_i characterizes the importance of the indicator in the system, and the larger the centrality, the more important the indicator is. Finally, from the centrality M_i , the subjective weights assigned by the DEMATEL method can be derived:

$$w_j = \frac{M_i}{\sum_{i=1}^n M_i} \quad (9)$$

where w_j denotes the weight of each indicator.

4.2. Entropy Weight Method Objective Weighting

The concept of entropy was first introduced into the field of information theory by Claude Elwood Shannon and has since been employed extensively in a range of disciplines, including engineering technology, social economy, and other fields. The entropy weight method is capable of determining the objective weight in accordance with the magnitude of the variability observed in line loss indicators. In general, a smaller information entropy value for an indicator indicates a greater degree of variability, whereby the indicator provides more information and plays a more significant role in the comprehensive evaluation. Consequently, the weight assigned to the indicator is also greater. Conversely, an indicator with a larger information entropy will exhibit a smaller degree of variability in its value. This indicates a reduction in the amount of information provided, which subsequently limits its capacity to contribute to the comprehensive evaluation of online losses and results in a reduction in the assigned weight [27].

$$e_j = -k \sum_{i=1}^n p_{ij} \ln(p_{ij}), j = 1, \dots, m \quad (10)$$

In the formula, e_j represents the entropy value of the j th index, and the p_{ij} sample value accounts for the proportion of the index.

Where n denotes the number of indicators, e_j denotes the entropy value of the j th indicator, and p_{ij} the weight of the sample value of the indicator.

$$w_j = \frac{1 - e_j}{\sum_{j=1}^m e_j}, j = 1, \dots, m \quad (11)$$

where w_j denotes the weight of each indicator.

4.3. Portfolio Evaluation Analysis Based on DEMATEL-Entropy Weight-TOPSIS

The DEMATEL method is a methodology proposed for screening the main elements of complex systems and simplifying the process of analyzing their structure. This method makes full use of the experience and knowledge of experts to deal with complex problems and is especially effective for those systems with uncertain element relationships. However, DEMATEL evaluation tends to be more subjective by experts and is easily influenced by personal thoughts, so the Entropy-Weight objective assignment method is proposed. The Entropy-Weight method can objectively reflect the differences between different indicators and the degree of importance, and the calculation results are more accurate, but it needs the indicators to conform to a normal distribution as much as possible, and it is not good for dealing with individual extreme values. At the same time, the indicators are not completely independent from each other, and the correlation between the indicators cannot be taken into account. Therefore, this paper will analyze the connotation of low carbon competitiveness based on the comprehensive use of a variety of evaluation ideas, the combination of subjective and objective evaluation methods, and the combination of multiple evaluation methods so that the subjective and objective evaluation methods can complement each other; the competitiveness of the regional line loss is a comprehensive, scientific, and objective evaluation, to ensure that the indicator weights not only truly reflect the objectivity of the data but also ensure that the indicator system has a predictive nature.

The TOPSIS (Technique for Order Preference by Similarity to an Ideal Solution) method is a sort of ordering method close to an ideal solution; it is based on a limited number of evaluation objects and the degree of proximity to the idealized target for the order is a commonly used and effective evaluation method in multi-objective decision-making analysis [28]. It is because the method can be more objective to the evaluation object to carry out scientific and reasonable sorting, so it is widely used by scholars to evaluate the sample data, but the defect of applying the general TOPSIS method is that it does not take into account the weights of the indicators, and it adopts a one-dimensional qualitative approach

to average the weights of the evaluation indicators to sort the evaluation object, which will lead to a certain research bias between the evaluation results and the objective actual values. This paper will be based on the above subjective and objective combination of calculated weight results, using the weighted TOPSIS method to construct a DEMATEL-Entropy Weight-TOPSIS evaluation method based on the Nanning City line loss competitiveness of the calendar year score calculation; the specific calculation process is as follows.

Same as the trending and indicator dimensionless, for both positive and negative indicators, the normalization formula is

$$a_{ij} = (x_{ij} - \min x_{ij}) / (\max x_{ij} - \min x_{ij}) \quad (12)$$

Calculation of the distances D_i^+ and D_i^- from the best and worst options for each evaluation object.

$$D_i^+ = \sqrt{\sum_{j=1}^m w_j (a_{ij} - a_{ij}^+)^2} \quad (13)$$

$$D_i^- = \sqrt{\sum_{j=1}^m w_j (a_{ij} - a_{ij}^-)^2} \quad (14)$$

Calculate the proximity of each evaluation object to the optimal and worst options: C_i :

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (15)$$

5. Construction of a Mapping of Management Cases for Comprehensive Line Loss Evaluation

5.1. Line Loss Indicator Model Based on Principal Component Analysis

The Guangxi Power Grid Company fulfils the stipulations set forth by Southern Power Grid Company's Line Loss Management Measures. In alignment with the tenets of comprehensive coverage, collectability, and efficacy in loss reduction, the company has divided the evaluation process into four distinct phases: planning loss reduction, management loss reduction, operational loss reduction, and technology loss reduction. This comprehensive approach encompasses the entirety of the line loss management process. Line loss management is a comprehensive indicator that reflects the level of high-quality development of the power grid. It involves a range of disciplines, including grid planning and construction, technology, operation and marketing, and other professions. In essence, it can be considered a "comprehensive physical examination" of the power grid. The causes of "abnormal line loss" in power grid equipment may be attributed to a multitude of factors, including the age of the lines, the length of the power supply radius, low voltage, heavy overload, and unstable trend operation mode. These factors often necessitate a significant investment of resources to investigate individually, with the aim of mitigating the risks posed to the power grid. The traditional method of calculating the line loss rates involves the use of meter readings. The traditional method of calculating line loss rates is based on meter readings taken over a period of time that may not be representative of the actual situation. This can result in significant monthly fluctuations, making it difficult to draw accurate conclusions and prescribe effective solutions.

In order to demonstrate the practicality of the indicator selection method proposed in this paper, the line loss indicator data of a city-level power grid enterprise in the Southern Power Grid will be used as an illustrative example. In order to ascertain the comprehensive line loss rate and the industry's evaluation of power grid enterprise line loss, a composite evaluation analysis method is selected for the calculation. The example employs statistical calculation data of 104 power supply enterprises' line loss reduction impact indicators, with each sample comprising 37 indicators from four principal categories: planning reduction, management reduction, operational reduction, and technical reduction. The influence of

the initial 13 kernel principal components is illustrated in Figure 6. From the figure, it can be observed that as the Gaussian kernel function's σ increases, the contribution rate of the principal components also increases. When $\sigma = 9$, the contribution rate of the principal components increases and becomes stable, indicating that the dimensionality reduction effect of the input data set space is effective and that the overfitting phenomenon is less likely to occur.

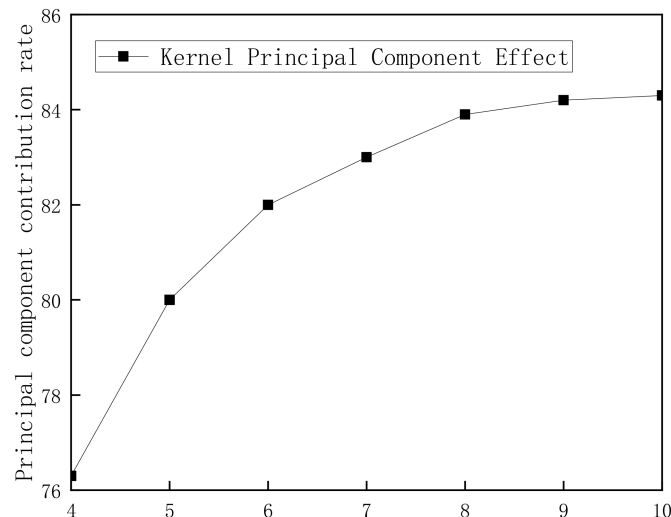


Figure 6. Impact diagram of kernel principal components.

In conclusion, the application of the Gaussian kernel function parameter with a standard deviation of 9 in the kernel principal component analysis (KPCA) method has been found to yield the most effective reduction in the dimensionality of the input dataset. The cumulative contribution rate of the 37 principal components is 100.00%. The initial 13 principal components account for 71.211% of the total contribution. When the cumulative contribution rate exceeds 70%, the higher the variance percentage corresponding to each extracted factor, the larger the eigenvalue of that factor, and the greater its role in explaining the original variables. In particular, the eigenvalues decrease in magnitude from the 13th factor onwards, indicating that these factors have a diminished capacity to explain the original variables and can thus be excluded from further consideration. In light of the aforementioned analysis, it can be posited that extracting the initial 13 factors is an appropriate course of action for the multiple detection indicators of line loss. This approach ensures an adequate explanation of the original variables while avoiding the introduction of superfluous factors, thereby rendering the analysis more concise and efficient. The eigenvalues and contribution rates of the KPCA at the optimal kernel parameter are presented in Table 1 for reference. The thirteen indicators for evaluating power loss management, together with their corresponding parameters, can be found in Supplementary Materials File S1. The equivalent parameters for the metrics enumerated in Table S1 of the Supplementary Materials correspond to the various indicators for power loss management. The equivalent parameters, which are abbreviated as indicators in the subsequent Supplementary Materials, serve two purposes. Firstly, they provide a set of standardized evaluation tools for power loss management. Secondly, they allow these parameters to be incorporated into a comprehensive evaluation model. The data types included in Supplementary Materials Table S2 encompass equipment, operational, capacity, and topology data. These data types not only provide a rich source of information, but also serve to establish a robust foundation for the weight analysis in the loss management assessment. As illustrated in the parallel coordinate charts in Figures S1 and S2 of the Supplementary Materials, the discrepancies between the data points in multiple dimensions are readily apparent. Nevertheless, the considerable volume of data precludes the direct generation of indicator weights and comprehensive evaluations. It is therefore necessary to undertake further analysis of the data in

order to determine the specific weight of each indicator in the comprehensive evaluation, with a view to more accurately reflecting the degree of influence of each indicator on the efficiency of power loss management.

Table 1. Eigenvalues and variance contribution of the principal component analysis method.

Ingredients	Amount to	Explanation of Total Variance Percentage of Variance	Cumulative %
Main variable capacity-to-load ratio	3.858	10.928	10.928
The pass rate of reactive power configuration of substation	2.933	8.426	19.354
Unplanned outage rate of distribution lines	2.854	7.712	27.066
Public change outage rate	2.448	6.617	33.683
The proportion of old low-voltage energy meters	2.130	5.758	39.441
The proportion of automatic meter reading	1.936	5.232	44.673
Data acquisition integrity rate of four types of terminals	1.870	5.053	49.726
The rate of emergency repair orders for hundreds of households	1.651	4.461	54.187
Electricity sales growth rate	1.550	4.189	58.376
The success rate of automatic execution of fee control	1.346	3.637	62.013
Power factor pass rate	1.174	3.173	65.186
Availability rate of substation reactive power compensation device	1.167	3.155	68.341
Energy-saving main variable ratio	1.062	2.870	71.211

The cumulative variance contribution rate of the 13 principal components reaches 71.211%, representing a proportion exceeding 71.211% of the characteristic information of the line loss indicators. The line loss influencing factors identified through kernel principal component analysis (KPCA) can be classified into four principal categories. Planning dimension indicators account for 37.470% of the characteristic information, management dimension indicators account for 32.762%, operational dimension indicators account for 14.286%, and technical dimension indicators account for 1.482%, for a total of 13 indicators. The feature extraction of line loss based on KPCA retains the features that contribute the most to the variance in the dataset. The replacement of 37 indicators with 13 has the effect of greatly reducing the computational load, which is a scientific evaluation of the objective phenomenon.

5.2. Data Analysis and Experimental Platform

There are as many as 3.05 million local municipal users, and the traditional customer power consumption accurate portrait and collection management is a very difficult and long cycle. This paper will be based on the basic data from 2014 to 2017 in Nanning City, using the combination of DEMATEL-Entropy Weight-TOPSIS evaluation method to carry out empirical research on the whole process of line loss management evaluation system. Based on the above subjective and objective combination of weight determination method, this paper will be in accordance with the scientific, objective, and reasonable evaluation ideas, on top of the analysis of specific data on the basis of the expert assignment of weights, to ensure that the design of the indicator system in the determination of the weight of the indicators is scientific, the weight of the indicator data is objective and reasonable, so that the subjective evaluation method of the expert assignment of the weight obtains W , W_i , the objective evaluation method of the weight W' , W'_i , the weights obtained.

According to the weight ratios $W:W' = 1:1$ and $W_i:W'_i = 1:1$, the weights of the evaluation indicators in the evaluation indicator system are calculated again.

5.3. Ablation Experiment Based on DEMATEL-Entropy Weight-TOPSIS Whole Process Combination Evaluation Method

In order to ascertain the veracity of each component of the prediction model DEMATEL-Entropy Weight-TOPSIS proposed in this paper, ablation experiments have been conducted on this model. Two ablation experiments are conducted, the first of which removes the DEMATEL structure, while the second removes the Entropy Weight. This paper participates in the ablation prediction comparison and evaluation methods, namely the DEMATEL-TOPSIS evaluation and the Entropy Weight-TOPSIS evaluation. These evaluation methods retain the distance between each scheme and the ideal solution in order to compare the

advantages and disadvantages of the scheme. Additionally, they facilitate a horizontal comparison of the analysis, allowing for the impact of each line loss indicator on the enterprise to be assessed.

The incorporation of entropy weighting in the ablation experiment serves to enhance the limitations inherent to the conventional human-subjective assignment of the TOPSIS method. Concurrently, it circumvents the potential distortion of the evaluation outcomes resulting from the influence of numerous human variables. The TOPSIS method does not necessitate a vast number of evaluation samples. It is capable of comprehensively and objectively reflecting the advantages and disadvantages of the various evaluation schemes through the information provided by the original data.

In order to address the impact of the interaction between the comprehensive assessment indicators of line loss on the system assessment, the DEMATEL method was employed to elucidate the relationship between the indicators. A total of 50 experts with a wealth of theoretical knowledge and field experience were invited to distinguish the indicators into cause factors and result factors. The degree of interaction between the various risk factors was scored in accordance with the rules of the DEMATEL analysis of influencing factors, as illustrated in Table 2. The scoring of the thirteen indicators and the comprehensive impact matrix are presented in Supplementary Materials File S2. The parameters set forth in the table header of Supplementary Materials File S2 are analogous to those delineated in Supplementary Materials Table S1. The approach taken in Supplementary Materials Table S3 is based on the established model of the factors that influence power loss data. By consulting relevant literature and experts, the influencing factors are evaluated and assigned a score on a scale of 0 to 4. This scale is used to classify the direct influence between the influencing factors, with 0 indicating no influence, 1 indicating a small influence, 2 indicating general influence, 3 indicating a large influence, and 4 indicating a very large influence. The corresponding scores between two factors indicate the degree of influence exerted by one factor on the other. The final values in the direct influence matrix are determined based on the relationships between the various influencing factors as evidenced in a substantial corpus of documents, thereby yielding the direct influence matrix. The centrality–influence diagram of each influencing factor in loss management is illustrated in Supplementary Materials Figure S3. A higher centrality value indicates a greater importance of the influencing factor in comparison to other factors. The term “influence degree” is employed to signify the extent to which a specific influencing factor exerts influence over other influencing factors. In the event that the influence degree is negative, this indicates that the influencing factor is susceptible to being affected by other factors. As illustrated in Supplementary Materials Figure S3, both the centrality and the influence are positive for Indicator 11, indicating that this indicator is significantly associated with other indicators and is more likely to affect other indicators.

The data pertaining to the line loss management of the Nanning City project were selected as a representative sample for the purpose of scoring the project’s line loss management level indicators. The degree of information redundancy was calculated, and the weights were determined using the entropy weight method in accordance with the aforementioned degree of information redundancy. The weights of each index within the comprehensive evaluation index system for line loss management are presented in Table 3.

Table 2. Influencing factors of fuzzy DEMATEL analysis.

Target	Impact f;	Affected e;	Center Cloud	Reason y;	Factor Attributes
Main variable capacity-to-load ratioU11	6.6453	5.7049	12.3502	0.9404	Cause factors
The pass rate of reactive power configuration of substationU12	6.3282	7.7028	14.0310	−1.3746	Result factors
Unplanned outage rate of distribution linesU13	4.9426	4.8259	9.7685	0.1167	Cause factors
Public change outage rateU14	7.1778	7.6720	14.8498	−0.4942	Cause factors
The proportion of old low-voltage energy metersU21	7.5627	8.0082	15.5709	−0.4455	Result factors
The proportion of automatic meter readingU22	7.1885	8.1637	15.3522	−0.9752	Result factors
Data acquisition integrity rate of four types of terminalsU23	7.4075	7.5046	14.9121	−0.0972	Cause factors
The rate of emergency repair orders for hundreds of householdsU24	6.4469	7.1053	13.5522	−0.6585	Cause factors
Electricity sales growth rateU25	7.1007	6.8508	13.9515	0.2499	Result factors
The success rate of automatic execution of fee controlU26	6.0915	5.7732	11.8647	0.3184	Result factors
Power factor pass rateU31	6.4481	5.1844	11.6325	1.2636	Result factors
Availability rate of substation reactive power compensation deviceU41	6.2500	4.7670	11.0170	1.4830	Result factors
Energy-saving main variable ratioU42	7.7616	8.0884	15.8500	−0.3269	Cause factors

Table 3. Entropy weight analysis of the influencing factors.

First-Level Indicators	First-Level Weight	Secondary Indicators	Second-Level Weight	Comprehensive Weight
Planned loss reduction	0.2532	Main variable capacity-to-load ratio	0.0366	0.0093
		The pass rate of reactive power configuration of substation	0.2476	0.0627
		Unplanned outage rate of distribution lines	0.2117	0.0536
		Public change outage rate	0.5041	0.1276
Manage loss reduction	0.4792	The proportion of old low-voltage energy meters	0.2866	0.1373
		The proportion of automatic meter reading	0.0131	0.0063
		Data acquisition integrity rate of four types of terminals	0.2866	0.1373
		The rate of emergency repair orders for hundreds of households	0.2460	0.1179
		Electricity sales growth rate	0.1678	0.0804
Operation loss reduction	0.1377	The success rate of automatic execution of fee control	0.5006	0.1340
		Power factor pass rate	0.4356	0.1166
Technical loss reduction	0.1299	Availability rate of substation reactive power compensation device	0.0136	0.0037
		Energy-saving main variable ratio	0.0502	0.0134

5.4. Comparison of Comprehensive Line Loss Evaluation Results under Ablation Evaluation Methods and Post-Hoc Tests

The standardized raw data presented in Tables 2 and 3, in conjunction with the combined index weights derived from the fuzzy DEMATEL and entropy weight methods, were employed to calculate the Euclidean distance between the positive and negative ideal solutions of the line loss abatement management capability utilizing Equation (12). Subsequently, the target layer affinity of the line loss management capability of the 11 electric power enterprises in 2014 was determined through the application of Equation (13), and the results were ranked in accordance with the magnitude of the affinity, as illustrated in Table 4. The results are presented in Table 4. The weighted normalization matrices of the DEMATEL-TOPSIS and Entropy TOPSIS evaluation methods for line loss management in Nanning City in 2014 are provided in Supplementary Materials File S3. Among the methods presented, Tables S4 and S5 of the Supplementary Materials DEMATEL-TOPSIS evaluation method employs the DEMATEL approach to ascertain the direct correlation between indicators. Subsequently, the TOPSIS method is utilized for the ultimate evaluation, disregarding the weight distribution determined by the entropy weight method. As can be observed from the weighted normalized matrix, the weights of certain indicators (such as C4 and C12) may be elevated, indicating that these indicators exert a greater influence within the evaluation model and contribute to the final evaluation results to a greater extent. The Entropy-TOPSIS evaluation method detailed in Tables S6 and S7 of the Supplementary Materials represents a contrasting approach to that described in the preceding section. This method employs the entropy weighting approach solely for determining the weight of the indicators, subsequently evaluating them using the TOPSIS method, eschewing the DEMATEL method. Both methods construct corresponding weighted and normalized matrices and are applied to the same data set of power losses. The matrices illustrate the distribution of weights for each indicator across the various methods. A comparison of the evaluation results obtained from the three methods allows for the observation of the specific impact of each component on the accuracy of the evaluation. The results of the ablation experiments serve to further confirm the effectiveness of the DEMATEL method in elucidating the intricate relationships between indicators, as well as the role of

the entropy weighting method in ensuring the objectivity of the weights attributed to the evaluation indicators.

Table 4. DEMATEL-Entropy Weight-TOPSIS model ablation study scores.

District	EW-TOPSIS	Rankings	D-TOPSIS	Rankings	EW-D-TOPSIS	Rankings
NN	0.5856	9	0.5852	4	0.5854	8
LZ	0.5854	12	0.5835	9	0.5846	10
GL	0.5857	4	0.5844	6	0.5855	5
WZ	0.5857	6	0.5843	7	0.5855	6
BH	0.5857	5	0.5856	2	0.5857	3
FCG	0.5855	10	0.5790	11	0.5843	11
QZ	0.5858	2	0.5856	3	0.5857	2
GG	0.5858	3	0.5848	5	0.5856	4
YL	0.5857	7	0.5837	8	0.5854	7
BS	0.5854	11	0.5769	12	0.5840	12
HZ	0.5851	14	0.5750	13	0.5828	13
HEC	0.5858	1	0.5858	1	0.5858	1
LB	0.5851	13	0.5696	14	0.5825	14
CZ	0.5856	8	0.5795	10	0.5849	9

As illustrated in Table 4, the investigation was conducted utilizing the rank correlation coefficient. The rank correlation coefficient, also referred to as the Spearman correlation coefficient, is a statistical measure derived by ordering the sample values of the two variables in accordance with their magnitude and replacing the original data with the rank of the sample values.

$$p = 1 - \frac{6}{n(n^2 - 1)} \sum_{i=1}^n d_i^2 \quad (16)$$

The t -test for the rank correlation coefficient ρ was performed using the following equation for $n > 8$. The test statistic is

$$t = \sqrt{n - 2p} / \sqrt{1 - p^2} \quad (17)$$

If $t \leq t_{n/2}(n - 2)$, it can be assumed that the results under the ablation evaluation method are not closely related to those analyzed; otherwise, the results of the combined evaluation are considered to be closely related to those of the original method.

As a result, the rank correlation coefficients between the original combination evaluation results and the DEMATEL-TOPSIS method and Entropy Weight-TOPSIS method are calculated to be 0.9512, 0.8593, and 0.6440, respectively; the corresponding t -values are 16.6018, 14.0463, and 13.5327, respectively; and the significance level of $\alpha = 0.05$ corresponding to the critical value of the test is $t_{\alpha/2}(n - 2) = t_{0.025}(29) = 2.045$. It is easy to judge that the results of DEMATEL-Entropy Weight-TOPSIS combination evaluation are closely related to the results of expert judgement.

The evaluation scores of line loss management capability of the sample companies in 2014 exhibited considerable variation. Through the application of clustering and stratification techniques, the top six companies in the stronger category of line loss management capability were assigned to Category A, the bottom four companies in the weaker category were designated as Category C, and the remaining companies were placed in the medium stratum. The four companies with the strongest line loss management capabilities are HEC, QZ, BH, and GG. HEC has the highest overall score for line loss management capability in 2014, indicating that the enterprise's overall line loss management capability is in a favorable position. YL has a line loss management capability score of 0.5854, which places it in the second tier. The line loss management capability situation is therefore in the average strength and the second tier of enterprises. The total number of cases constitutes 42.86 percent of the study sample. The company with the lowest overall score is LB, with

a score of 0.5825. This is a significant discrepancy from the management ability of the first-ranked enterprise, indicating that there is considerable potential for improvement in line loss management ability.

5.5. Research on the Whole-Process Evaluation System of Line Loss Management in a City of Guangxi District Based on DEMATEL-Entropy Weight-TOPSIS

The collection of indicator data is conducted in accordance with the established evaluation indicator system. The determination of the qualified value is based on a combination of national standards, industry standards or norms, and expert experience. Table 5 presents a comprehensive overview of the scores for each urban area in Guangxi, based on the combined evaluation conducted between 2014 and 2016. It also illustrates the scores for each secondary indicator.

Table 5. Topsis composite score and ranking results for each secondary indicator score table for all years.

	Overall Ranking			Planning for Loss Reduction			Managing Loss Reduction			Operational Loss Reduction			Technical Loss Reduction		
	14	15	16	14	15	16	14	15	16	14	15	16	14	15	16
NN	8	1	2	2	2	14	13	12	8	5	10	12	13	12	10
LZ	10	12	3	3	3	2	11	13	9	10	12	5	12	7	6
GL	5	6	5	11	1	7	8	14	7	1	9	9	6	3	12
WZ	6	14	13	5	5	6	9	10	10	4	7	8	1	5	5
BH	3	2	1	14	11	11	14	5	5	12	12	11	14	4	7
FCG	11	9	8	10	7	5	6	8	12	6	4	4	9	8	4
QZ	2	13	11	9	14	13	7	9	13	11	11	13	10	6	3
GG	4	11	9	7	13	12	3	6	6	8	5	7	4	2	2
YL	7	5	4	8	9	9	2	3	1	1	3	3	8	11	14
BS	12	7	14	6	12	1	5	1	14	12	12	14	1	14	9
HZ	13	4	12	1	4	8	12	11	4	12	1	1	11	10	8
HEC	1	10	6	4	8	3	10	7	11	9	8	10	1	1	1
LB	14	3	7	13	6	4	1	4	3	1	1	1	5	13	13
CZ	9	8	10	12	10	10	4	2	2	7	6	6	7	9	11

5.6. Comparative Analysis

The evaluation results for line loss management capability from 2014 to 2016 are classified into three categories. Those ranked in the top six are classified as Category A, indicating a high level of line loss reduction management capability. Those ranked in the bottom four are classified as Category C, indicating a low level of line loss management capability. The remaining results are classified as Category B, indicating a mid-range level of line loss management capability. For a detailed breakdown of the classification, please refer to Table 6.

Table 6. Overall ranking breakdown of line loss management capabilities, 2014–2016.

	Year 2014	Year 2015	Year 2016
Category A	HEC(1)QZ(2)BH(3) GG(4)GL(5)WZ(6)	NN(1)BH(2)LB(3) HZ(4)YL(5)GL(6)	BH(1)NN(2)LZ(3) YL(4)GL(5)HEC(6)
Category B	YL(7)NN(8)CZ(9) Average (10)LZ(11)	BS(7)CZ(8)FCG(9) Average (10)HEC(11)	LB(7)FCG(8) Average (9) GG(10)CZ(11)
Category C	FCG(12)BS(13) HZ(14)LB(15)	GG(12)LZ(13) QZ(14)WZ(15)	QZ(12)HZ(13) WZ(14)BS(15)

In the period spanning 2014 to 2016, both BH and GL municipalities were classified as Class A line loss management companies. Similarly, the three municipal corporations of NN,

HEC, and YL were classified as Class A line loss management companies in two of the years between 2014 and 2016. The two cities of BH and GL, which are popular tourist destinations, have a high level of economic development and relatively well-developed infrastructure and advanced power management technology, which contributes to a reduction in line loss. The two cities benefit from the government's policy support and regional development strategy, which may include the priority of grid construction and upgrading, as well as access to technical and financial support. It is plausible that the three urban areas of NN, HEC, and YL have undertaken grid construction and technological upgrading over the past two years. This may have led to an improvement in the transmission efficiency of the grid and a reduction in line loss, particularly for the two companies NN and YL, whose overall index rank and scores have exhibited a consistent upward trajectory. This suggests the presence of an effective line loss management model.

For a period of two years between 2014 and 2016, four municipalities, BS, HZ, QZ and WZ, were categorized as Category C line loss management companies. These areas are subject to national policy support and attention, particularly in regard to ethnic minority issues. They may also possess specific geopolitical advantages and challenges, notably in the context of poverty alleviation, regional development, and ethnic minority policies. Amongst these, QZ and WZ were initially classified as Category A for their line loss management in 2014. However, they were subsequently reclassified as Category C for two consecutive years from 2015 to 2016. Consequently, they may encounter similar economic development challenges, including the necessity to enhance infrastructure development, facilitate industrial upgrading, and improve the investment environment, in addition to other pertinent issues.

The remaining provinces and cities were classified as category B from 2014 to 2016, including LZ, FCG, GG, LB, and CZ. The findings of the study indicate a notable geographical variation in line loss management capacity within the Guangxi region. In terms of the overall ranking, the line loss management capability is stronger in tourist coastal areas and economically developed areas, weaker in ethnic minority and economically weaker areas, and the rest of the regions fall into the middle category.

In order to ascertain the reasons for the discrepancies in the overall ranking, the results of the ranking of the sub-items pertaining to line loss management ability were classified and subjected to analysis. The management loss reduction capability, which accounts for the largest number of indicators, is taken as an example here. The classification table of management loss reduction capability is shown in Table 7.

Table 7. Overall ranking breakdown of loss reduction capabilities in line loss management category, 2014–2016.

	Year 2014	Year 2015	Year 2016
Category A	LB(1)YL(2)GG(3) CZ(4)BS(5)FCG(6)	BS(1)CZ(2)YL(3) LB(4)BH(5)GG(6)	YL(1)CZ(2)LB(3) HZ(4)BH(5)GG(6)
Category B	QZ(7)GL(8) WZ(9)HEC(10)	HEC(7)FCG(8) QZ(9)WZ(10)	GL(7)NN(8) LZ(9)WZ(10)
Category C	LZ(11)HZ(12) NN(13)BH(14)	HZ(11)NN(12) LZ(13)GL(14)	HEC(11)FCG(12) QZ(13)BS(14)

The municipalities of LB, YL, GG, and CZ collectively demonstrate a robust capacity in the domain of line loss management, exhibiting consistently high levels across the three-year period from 2005 to 2007. Consequently, they are classified as A. In contrast, the municipality of BH exhibited a B classification in 2014 but subsequently demonstrated superior performance, achieving an A classification from 2015 to 2016. Consequently, it is also classified as A. BS was classified as A from 2014 to 2015 and is therefore also classified as A. QZ The municipal bureau's line loss management input capacity is insufficient, ranking 7th in 2014, and has subsequently declined, from 2015 to 2016, to 13th, belonging to category

B. QZ should continue to increase investment to enhance the local management technology input capacity. GL management input capacity is also insufficient, belonging to category B, but the ranking has risen. NN, HZ, and LZ are three municipal bureaus with weaker management capabilities in the area of line loss. It may be the case that these bureaus suffer from weaker management capacity, which may be related to their management planning and system. In order to enhance the effectiveness of line loss management, it would be beneficial to implement a more scientific and reasonable comprehensive evaluation system.

A comparison of the aforementioned case mapping visual analysis display with the existing comprehensive line loss management of Southern Power Grid during the same period reveals that knowledge mapping can provide managers with more comprehensive and accurate data support following the intelligent integration and analysis of data sources. In comparison with the conventional approach of analyzing data from a single source, the comprehensive evaluation methodology enables the integration of multiple data sources, thereby facilitating a more comprehensive understanding of the nature of the line loss problem among managers. The utilization of knowledge mapping enables the simultaneous updating of data and knowledge in real-time, thereby facilitating intelligent line loss evaluation management. This approach reduces the necessity for manual intervention, enhances management efficiency and accuracy, and significantly accelerates the determination of comprehensive evaluation management of line loss in comparison to traditional methods. Furthermore, the knowledge graph can leverage data mining and predictive analysis techniques to identify potential line loss risks and issue warnings in advance.

5.7. Research Discussion

This study proposes a comprehensive evaluation system for line loss based on a knowledge graph, which integrates kernel principal component analysis (KPCA) with the DEMATEL-Entropy Weight-TOPSIS combined evaluation method through a knowledge graph. In comparison to traditional deep learning models, this model integrates domain expert knowledge with statistical analysis techniques, thereby demonstrating notable advantages, particularly in the context of nonlinear and high-dimensional data.

Despite their efficacy in feature extraction and classification, deep learning models have inherent limitations in terms of model interpretability, as evidenced by the literature [20]. In contrast, this study enhances both the prediction accuracy and the interpretability of the results by incorporating a knowledge graph. As discussed in literature [22,23] regarding hybrid systems, this study also employs a mixed methodology, underscoring the significance of knowledge graphs in integrating multi-source data and reducing data redundancy. In comparison to the hybrid models based on fuzzy logic and rough sets referenced in literature [24,25], this model utilizes more sophisticated techniques, such as the Gaussian kernel function, to achieve enhanced performance in power system data analysis.

Concurrently, this model may encounter difficulties in terms of computational resources when dealing with large-scale and intricate power grid systems, a challenge analogous to that identified in literature [26]. The subsequent introduction of algorithm optimization and efficient data processing technologies can effectively mitigate this issue. In comparison to the models discussed in literature [27,28], this model may exhibit certain limitations with regard to real-time data processing and dynamic environmental adaptability. Further research will investigate the incorporation of more sophisticated machine learning techniques to augment the model's capacity for real-time performance and adaptability.

From a more profound perspective, the comprehensive evaluation system for line loss based on a knowledge graph strives to furnish a comprehensive array of line loss management solutions for power grid operators, policymakers, and engineers. Nevertheless, in practical applications, it is incumbent upon different customers to ensure the practical application of the proposed system based on clear and feasible recommendations.

A. It is recommended that operators integrate existing power grid data resources, including equipment operation data, user electricity usage data, and geographical

information data, and establish unified data standards to facilitate the construction and application of knowledge graphs. It is recommended that knowledge graphs and real-time data processing technologies be employed to establish dynamic monitoring systems, which will enable the real-time analysis of the operational status of the power grid and the timely detection and response to line loss anomalies. The development of predictive maintenance models based on knowledge graphs and machine learning technologies will facilitate the prediction of equipment failures and line loss trends, thereby reducing the incidence of unexpected power outages and associated maintenance costs.

- B. It is recommended that policymakers formulate relevant policies to encourage power grid companies to adopt advanced line loss management technologies, such as knowledge graphs and artificial intelligence, with the aim of improving the operational efficiency and reliability of the power grid. It is recommended that cooperation between the power industry and other relevant industries, such as information technology and data analysis, be promoted in order to facilitate joint development and application of innovative power grid management solutions.
- C. It is recommended that engineers receive training in the construction of knowledge graphs, data analysis, and the applications of artificial intelligence, in order to enhance their ability to apply new technologies. Concurrently, it is recommended that customized development be adopted and that line loss management models be developed that are adapted to the local environment. This should be performed in accordance with the characteristics of power grids in different regions, taking into account the differences between urban and rural power grids, for example.

It is also essential to consider the model's adaptability in different environments, such as urban and rural contexts. Urban power grids are characterized by intricate structural complexity and a high density of users, whereas rural power grids often encompass expansive areas with a dispersed population. It is essential that the model takes these differences into account. For instance, urban power grids require a greater focus on load management and the demand response, whereas rural power grids necessitate a stronger emphasis on infrastructure construction and maintenance. Additionally, the model should integrate environmental factors, such as climate, geography, and population distribution, in order to adapt to the specific conditions of different regions.

6. Conclusions

This paper constructs a comprehensive evaluation system of line loss based on a knowledge graph, and the main conclusions are as follows:

- A. A new comprehensive evaluation management knowledge map of line loss is constructed, which is divided into three modules to effectively target the existing line loss characteristics; it is more refined than the traditional map library and describes the analysis when selecting indicators in a more detailed way.
- B. A new comprehensive evaluation management system for line loss reduction is proposed based on the extraction of indicators from the knowledge map of line loss management, from which four categories and 13 types of indicators are selected to build a comprehensive evaluation model for line loss reduction. The model reacts more comprehensively to the correlation between indicators, achieves the accurate positioning of existing problems, and reduces the problem of the redundancy of similar line loss problems to increase the amount of calculation.
- C. A combined evaluation method based on DEMATEL-Entropy Weight-TOPSIS is proposed, and the reliability of the evaluation method is verified through ablation experiments. After the case study of line loss in Nanning City, the authenticity and reliability of the comprehensive evaluation system for evaluating the line loss is tested, and the strengths and weaknesses of the index system are judged more accurately, reducing the probability of misjudgment.

In the paper, the comprehensive evaluation and management system of line loss reduction is studied; with the increasingly large scale of the distribution network, the line loss situation is becoming more and more complex, and the line loss in the same area will also be due to cyclical changes in the characteristics and laws of the users' electricity consumption in the area, so it is necessary to further study to accurately grasp the real situation of the area's line loss.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/app14209462/s1>.

Author Contributions: Conceptualization, B.L. and W.W.; methodology, W.W.; software, W.W.; validation, B.L. and W.W.; formal analysis, W.W.; investigation, W.W.; resources, B.L.; data curation, B.L.; writing—original draft preparation, W.W.; writing—review and editing, W.W.; visualization, W.W.; supervision, B.L.; funding acquisition, B.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported in part by the Natural Science Foundation of Guangxi Province under grant 2020GXNSFAA297117.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data used in this study were obtained from China Southern Power Grid and have been licensed for sharing with other researchers upon request. Restrictions may apply to the availability of these data, which were used under license for this study and are not publicly available. Data are, however, available from the authors upon reasonable request and with permission from China Southern Power Grid.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Li, X.I.; Zhu, H.; Zhang, L. Review on China's National Carbon Market and Analysis for Commercial Development Potential of Carbon Capture, Utilization and Storage in China. In Proceedings of the 2022 IEEE PES Innovative Smart Grid Technologies—Asia (ISGT Asia), Singapore, 1–5 November 2022; pp. 804–807.
2. Anuradha, M.A.; Massoud, A. IEEE Vision for Smart Grid Controls: 2030 and Beyond. In Proceedings of the IEEE Vision for Smart Grid Controls: 2030 and Beyond, New York, NY, USA, 20 June 2013; pp. 1–168.
3. Amiri, S.S.; Rahmani, M.; McDonald, J.D. An Updated Review on Distribution Management Systems within a Smart Grid Structure. In Proceedings of the 2021 11th Smart Grid Conference (SGC), Tabriz, Iran, 7–9 December 2021; pp. 1–5.
4. Kor, Y.; Tan, L.; Musilek, P.; Reformat, M.Z. Integrating Knowledge Graphs into Distribution Grid Decision Support Systems. *Future Internet* **2024**, *16*, 2. [\[CrossRef\]](#)
5. Sabouhi, H.; Doroudi, A.; Fotuhi-Firuzabad, M.; Bashiri, M. Electrical Power System Resilience Assessment: A Comprehensive Approach. *IEEE Syst. J.* **2020**, *14*, 2643–2652. [\[CrossRef\]](#)
6. Kazemi, S.; Lehtonen, M.; Fotuhi-Firuzabad, M. *A Comprehensive Approach for Reliability Worth Assessment of the Automated Fault Management Schemes*; IEEE PES T&D: New Orleans, LA, USA, 2010; pp. 1–8.
7. Shreyash, N.K. Geo-Intelligent Architecture for Smart Grid Evolution: Addressing Contemporary Challenges through Spatial AI and Knowledge Integration. In Proceedings of the 2023 4th Asia Service Sciences and Software Engineering Conference (ASSE '23), Aizu-Wakamatsu, Japan, 27–29 October 2023; Association for Computing Machinery: New York, NY, USA, 2023; pp. 171–180.
8. Pahuja, V.; Wang, B.; Latapie, H.; Srinivasa, J.; Su, Y. A Retrieve-and-Read Framework for Knowledge Graph Link Prediction. In Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23), Birmingham, UK, 21–25 October 2023; Association for Computing Machinery: New York, NY, USA, 2023; pp. 1992–2002.
9. Elahi, E.; Anwar, S.; Shah, B.; Halim, Z.; Ullah, A.; Rida, I.; Waqas, M. Knowledge Graph Enhanced Contextualized Attention-Based Network for Responsible User-Specific Recommendation. *ACM Trans. Intell. Syst. Technol.* **2024**, *15*, 1–24. [\[CrossRef\]](#)
10. Xue, R.; Zhang, C.; Yan, H.; Lakhari, I.A.; Disasa, K.N.; Zhou, Y.; Li, J.; Wang, X.; Zhou, R.; Wang, B.; et al. Evaluating effect of micro-spray on tomatoes for resisting summer heat stress using a Fuzzy Borda combination evaluation model. *Environ. Exp. Bot.* **2024**, *218*, 105605. [\[CrossRef\]](#)
11. Jeon, H.; Lee, S.; Kim, H.; Soh, S.B.; Kim, S. Portfolio Evaluation with the Vector Distance Based on Portfolio Composition. *Mathematics* **2023**, *11*, 221. [\[CrossRef\]](#)
12. Amiri, M.; Balakrishnan, N.; Eftekharian, A. Hessian orderings of multivariate normal variance-mean mixture distributions and their applications in evaluating dependent multivariate risk portfolios. *Stat. Methods Appl.* **2022**, *31*, 679–707.

13. Liu, R.; Fu, R.; Xu, K.; Shi, X.; Ren, X. A Review of Knowledge Graph-Based Reasoning Technology in the Operation of Power Systems. *Appl. Sci.* **2023**, *13*, 4357. [\[CrossRef\]](#)
14. Chen, Q.; Li, Q.; Wu, J.; Mao, C.; Peng, G.; Wang, D. Application of knowledge graph in power system fault diagnosis and disposal: A critical review and perspectives. *Front. Energy Res.* **2022**, *10*, 988280. [\[CrossRef\]](#)
15. Chen, J.; Lu, G.; Pan, Z.; Yu, T.; Ding, M.; Yang, H. Research Review of the Knowledge Graph and its Application in Power System Dispatching and Operation. *Front. Energy Res.* **2022**, *10*, 896836. [\[CrossRef\]](#)
16. Xie, B.; Rashid, I.M.A. Research on the Application Mechanism of Enterprise Knowledge Management System Based on Computer Knowledge Graph Technology. In Proceedings of the 2023 International Conference on Internet of Things, Robotics and Distributed Computing (ICIRDC), Rio De Janeiro, Brazil, 29–31 December 2023; pp. 471–475.
17. Zheng, L.; Zheng, Z.; Chen, Y.; Qi, A. Environmental protection evaluation of new inorganic non-metallic building materials based on principal component analysis. *Int. J. Mater. Prod. Technol.* **2023**, *67*, 178–193. [\[CrossRef\]](#)
18. Wendt, J.D.; Wells, R.; Field, R.V.; Soundarajan, S. On data collection, graph construction, and sampling in twitter. In Proceedings of the 2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM), San Francisco, CA, USA, 18–21 August 2016; IEEE: Piscataway, NJ, USA, 2016; pp. 985–992.
19. Wu, X.; Li, G. Intelligent Agricultural Data Collection and Analysis System Based on Internet of Things. In Proceedings of the 2023 IEEE 2nd International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA), Changchun, China, 24–26 February 2023; IEEE: Piscataway, NJ, USA, 2023; pp. 506–509.
20. Le, L.; Hao, J.; Xie, Y.; Priestley, J. Deep Kernel: Learning Kernel Function from Data Using Deep Neural Network. In Proceedings of the 2016 IEEE/ACM 3rd International Conference on Big Data Computing Applications and Technologies (BDCAT), Shanghai, China, 6–9 December 2016; pp. 1–7.
21. Garniwa, I. Principal component analysis and cluster analysis for development of electrical system. In Proceedings of the 2017 15th International Conference on Quality in Research (QiR): International Symposium on Electrical and Computer Engineering, Bali, Indonesia, 24–27 July 2017; IEEE: Piscataway, NJ, USA, 2017; pp. 439–443.
22. Pei, Y. Linear principal component discriminant analysis. In Proceedings of the 2015 IEEE International Conference on Systems, Man, and Cybernetics, Hong Kong, China, 9–12 October 2015; IEEE: Piscataway, NJ, USA, 2015; pp. 2108–2113.
23. Xu, Z.; Wu, J. Comprehensive evaluation model of smart city economic management efficiency based on multidimensional data mining. In Proceedings of the 2021 International Conference of Social Computing and Digital Economy (ICSCDE), Chongqing, China, 28–29 August 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 92–95.
24. Cheng, X.; Yu, M.; Liu, M.; Huang, R.; Xie, L.; Tan, H. Research on comprehensive performance evaluation method of smart energy meter. In Proceedings of the 2018 3rd International Conference on Mechanical, Control and Computer Engineering (ICMCCE), Huhhot, China, 14–16 September 2018; IEEE: Piscataway, NJ, USA, 2018; pp. 450–455.
25. Wang, T.; Sun, Y.; Song, X.; Wang, Z.; Ren, Y.; Xu, W. Analysis on Causes of Train Operation Conflicts in High-Speed Railway Based on Fuzzy DEMATEL-ISM. In Proceedings of the 2020 Chinese Automation Congress (CAC), Shanghai, China, 6–8 November 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 2318–2323.
26. Bashardoost, P.; Nasirzadeh, F.; Mohtashemi, N.N. An integrated fuzzy-DEMATEL approach to project risk analysis. In Proceedings of the 2018 7th International Conference on Industrial Technology and Management (ICITM), Oxford, UK, 7–9 March 2018; IEEE: Piscataway, NJ, USA, 2018; pp. 411–416.
27. Wang, Y.; Tian, L.; Chen, Z. A Reputation Bootstrapping Model for E-Commerce Based on Fuzzy DEMATEL Method and Neural Network. *IEEE Access* **2019**, *7*, 52266–52276. [\[CrossRef\]](#)
28. Qi, J.; Liu, J.; Liu, Z.; Wang, K. Evaluation Method of Power Channel Operation Quality Based on Entropy Weight Method. In Proceedings of the 2023 3rd International Conference on New Energy and Power Engineering (ICNEPE), Huzhou, China, 24–26 November 2023; IEEE: Piscataway, NJ, USA, 2023; pp. 1014–1017.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.